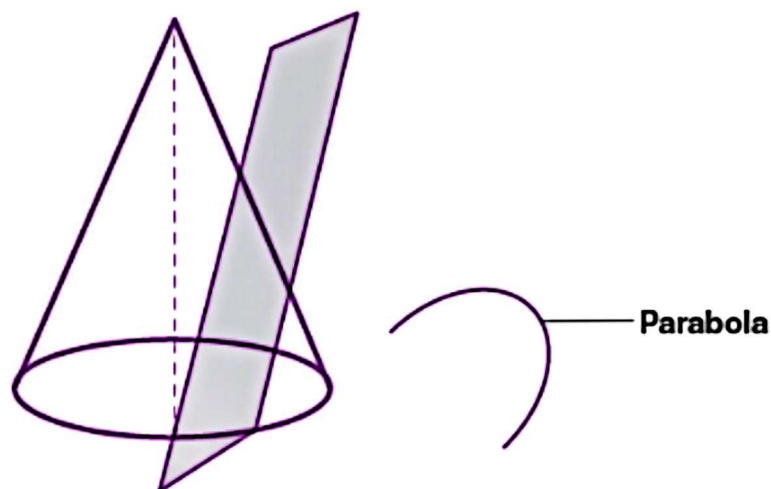


What Is a Parabola?

The section of a right circular cone by a plane parallel to a generator of the cone is a **parabola**. It is a locus of a point, which moves so that the distance from a fixed point (focus) is equal to the distance from a fixed line (directrix).

- A fixed point is called the focus
- A fixed line is called the directrix



Standard Equation of Parabola

In general, if the directrix is parallel to the y -axis in the standard equation of a parabola is given as

$$y^2 = 4ax$$

If the parabola is sideways, i.e., the directrix is parallel to the x -axis, the standard equation of a parabola becomes

$$x^2 = 4ay$$

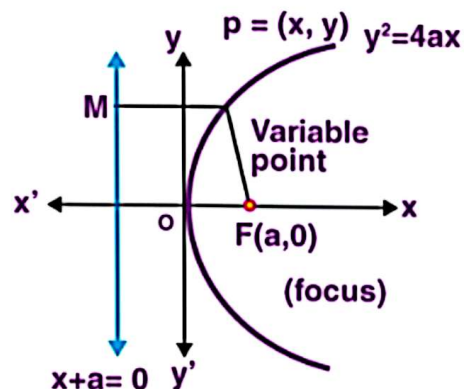
Apart from these two, the equation of a parabola can also be $y^2 = -4ax$ and $x^2 = -4ay$, if the parabola is in the negative quadrants. Thus, the four equations of a parabola are given as



1. $y^2 = 4ax$
2. $y^2 = -4ax$
3. $x^2 = 4ay$
4. $x^2 =$

Parabola Equation Derivation

In the above equation, "a" is the distance from the origin to the focus. Below is the derivation for the parabola equation. First, refer to the image given below.



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Since the eccentricity of the parabola is 1,
 $PM / PF = 1$

$$PM = PF$$

Using the distance formula,

$$\sqrt{(x-a)^2 + y^2} = \left| \frac{x+a}{1} \right|$$
$$(x-a)^2 + y^2 = (x+a)^2$$

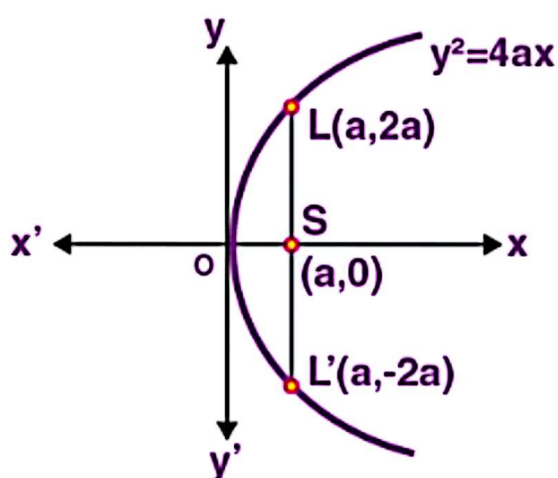
$$y^2 = 4ax$$

⇒ **Standard equation of Parabola**



Latus Rectum of Parabola

The latus rectum of a parabola is the chord that passes through the focus and is perpendicular to the axis of the parabola.



$$LSL' = \sqrt{(a-a)^2 + (2a+2a)^2}$$

$$= \sqrt{16a^2}$$

$$= 4a \text{ (Length of latus rectum)}$$

Note: – Two parabolas are said to be equal if their latus rectum is equal.

Parametric Co-ordinates of Parabola

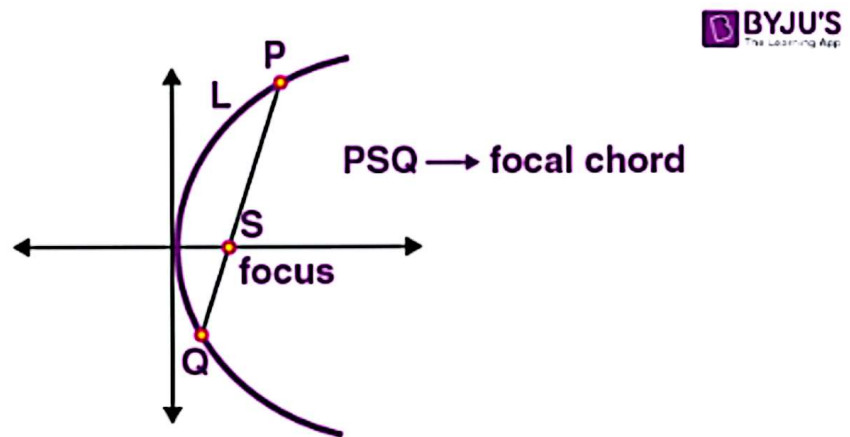
Consider a parabola, $y^2 = 4ax$. Now, to represent the coordinates of a point on the parabola, the easiest form will be $x = at^2$ and $y = 2at$ as for any value of "t", the coordinates $(at^2, 2at)$ will always satisfy the parabola equation, i.e., $y^2 = 4ax$.

So, $x = at^2$ and $y = 2at$ are called the parametric equations of the parabola $y^2 = 4ax$ (where t is the parameter).

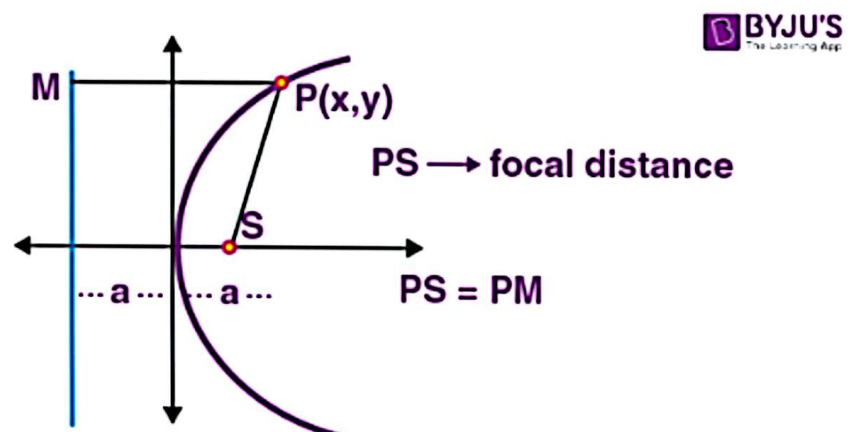
Parametric equations of the parabola $x^2 = 4ay$ are $x = 2at$, $y = at^2$

Focal Chord and Focal Distance

Focal chord: Any chord that passes through the focus of the parabola is the focal chord of the parabola. It cuts the parabola at two distinct points.



Focal Distance: The distance of any point, $p(x, y)$ on the parabola from the focus, is the focal distance. PS is the focal distance in the above figure.



General Equations of Parabola

The general equation of a parabola is given by $y = a(x - h)^2 + k$ or $x = a(y - k)^2 + h$. Here, (h, k) denotes the vertex.

$y = a(x - h)^2 + k$ is the regular form

$x = a(y - k)^2 + h$ is the sideways form

Position of a point with respect to the parabola

For parabola $y^2 = 4ax$, the position of the point $P(x_1, y_1)$ depends on the following conditions.

$$S_1 = y_1^2 - 4ax_1$$

If $S_1 > 0$, point P lies outside the parabola.

If $S_1 < 0$, point P lies inside the parabola.

If $S_1 = 0$, point P lies on the parabola.



The intersection of a straight line with the parabola

Let $y^2 = 4ax$ be the parabola and $y = mx + c$ is the straight line.

A line can meet a parabola at most two points.

$$(mx + c)^2 - 4ax = 0$$

$$m^2x^2 + x(2mc - 4a) + c^2 = 0.$$

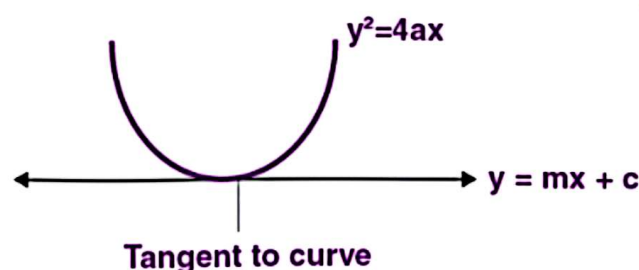
Here discriminant, $D = (2mc - 4a)^2 - 4m^2c^2$

If the discriminant is zero, then the line is tangent to the parabola.

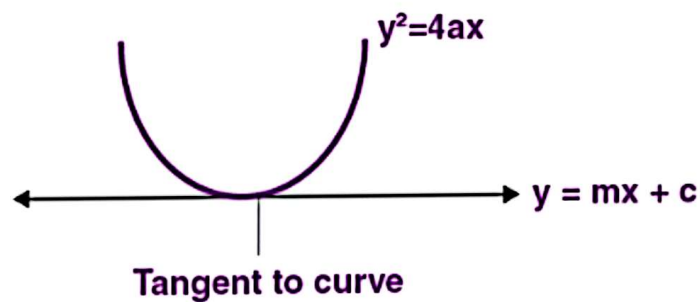
If the discriminant is less than zero, the line does not meet the parabola at all.

If the discriminant is greater than zero, the line meets the parabola at two distinct points.

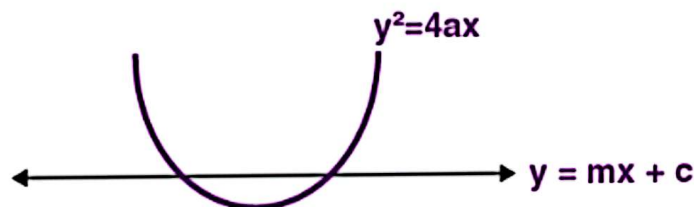
$D = 0$: Straight line is the tangent to the parabola.



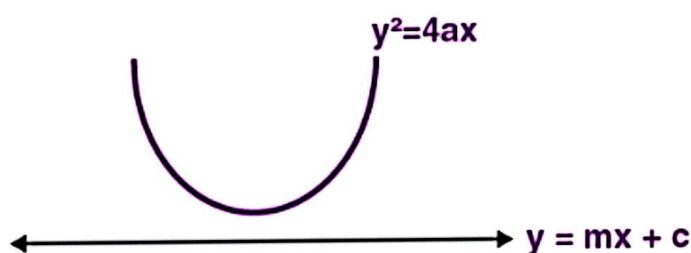
$D = 0$: Straight line is the tangent to the parabola.



$D > 0$: Straight line intersect the parabola at two points.



$D < 0$: Straight line does not touch the parabola.



Equation of Tangent to a Parabola

A line which touches the parabola exactly at one point is called the **tangent to a parabola**.

Point Form

The equation of tangent to parabola $y^2 = 4ax$ at (x_1, y_1) is $yy_1 = 2a(x+x_1)$.

The point of contact of the tangent is (x_1, y_1) .

Parametric Form

The equation of the tangent to the parabola $y^2 = 4ax$ at $(at^2, 2at)$ is $ty = x + at^2$.

The point of contact of tangent is $(at^2, 2at)$

Slope Form

If m is the slope of the tangent to the parabola $y^2 = 4ax$, then the equation of tangent is $y = mx + a/m$.

The point of contact of tangent is $(a/m^2, 2a/m)$



Pair of tangents from (x_1, y_1) external points

Let $y^2 = 4ax$ be the parabola and $P(x_1, y_1)$ be the external point, then the equation of tangents is given by

$$SS_1 = T^2$$

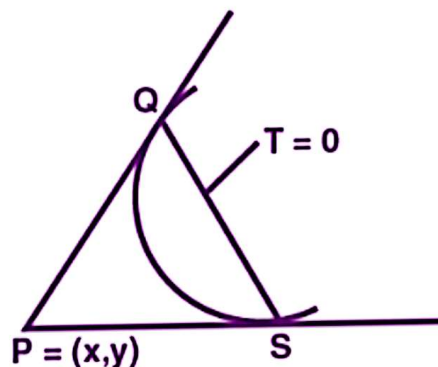
$$S = y^2 - 4ax, S_1 = y_1^2 - 4ax_1$$

$$T = yy_1 - 2a(x + x_1).$$

Chord of contact

Equation of chord of contact of tangents from a point $p(x_1, y_1)$ to the parabola $y^2 = 4ax$ is given by $T = 0$

$$\text{i.e., } yy_1 - 2a(x + x_1) = 0$$



The equation of QS is $T = 0$

Normal to the Parabola

The line perpendicular to the tangent of the parabola at the point of contact is the normal of a parabola.

The [equation of normal to a parabola](#) can be given in point form, parametric form and slope form.

Point Form

The equation of normal to the parabola $y^2 = 4ax$ at (x_1, y_1) is given by $y - y_1 = (-y_1/2a)(x - x_1)$.

Slope Form

The equation of normal to the parabola $y^2 = 4ax$ at $(am^2, -2am)$ is given by $y = mx - 2am - am^3$.

The point of contact is $(am^2, -2am)$.

Parametric Form

The equation of normal to the parabola $y^2 = 4ax$ at point $(at^2, 2at)$ is given by $y = -tx + 2at + at^3$.

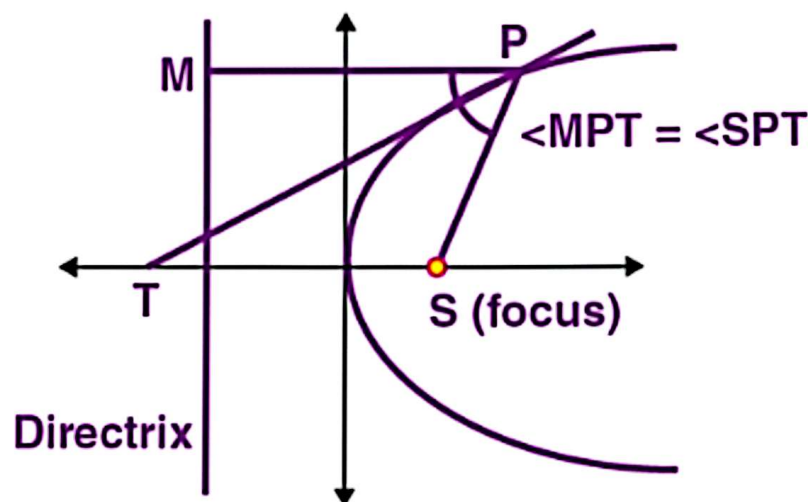
Important Properties of Focal Chord

1. If chord joining $P = (at_1^2, 2at_1)$ and $Q = (at_2^2, 2at_2)$ is focal chord of parabola $y^2 = 4ax$, then $t_1t_2 = -1$.
2. If one extremity of a focal chord is $(at_1^2, 2at_1)$, then the other extremity $(at_2^2, 2at_2)$ becomes $(a/t_1^2, -2a/t_1)$.
3. If the point $(at^2, 2at)$ lies on parabola $y^2 = 4ax$, then the length of focal chord PQ is $a(t + 1/t)^2$.
4. The length of the focal chord, which makes an angle θ with a positive x-axis, is $4a \operatorname{cosec}^2 \theta$.
5. Semi latus rectum is a harmonic mean between the segments of any focal chord.
6. Circle described on focal length as diameter touches tangent at the vertex. The circle, described on any focal chord of a parabola as diameter, touches the directrix.



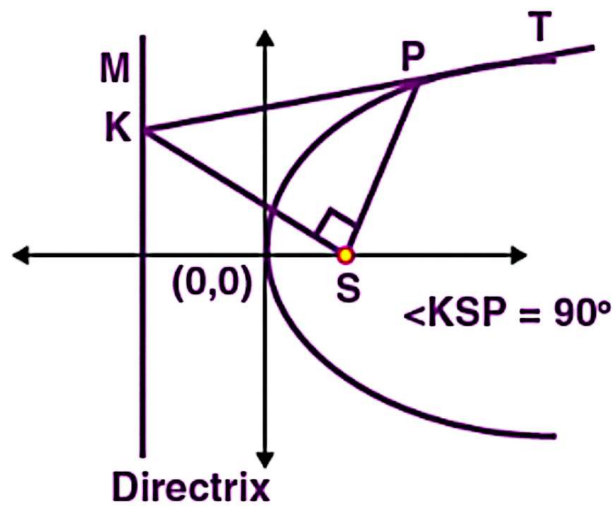
Important Properties of Focal Chord, Tangent and Normal of Parabola

- The tangent at any point P on a parabola bisects the angle between the focal chord through P and the perpendicular from P on the directrix.

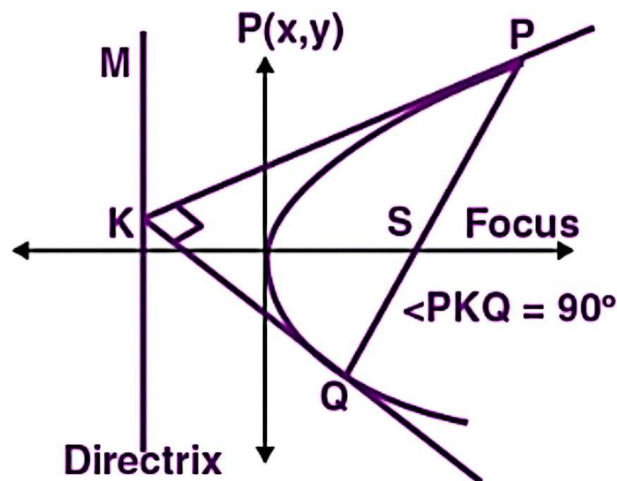


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- The portion of a tangent to a parabola cut off between the directrix and the curve subtends a right angle at the focus.



- Tangents at the extremities of any focal chord intersect at right angles on the directrix.



- The perpendicular drawn from a focus upon any tangent of a parabola lies on the tangent at the vertex.



Four Common Forms of a Parabola

Form:	$y^2 = 4ax$	$y^2 = -4ax$	$x^2 = 4ay$	$x^2 = -4ay$
Vertex	(0, 0)	(0, 0)	(0, 0)	(0, 0)
Focus	(a, 0)	(-a, 0)	(0, a)	(0, -a)
Equation of the directrix	$x = -a$	$x = a$	$y = -a$	$y = a$
Equation of the axis	$y = 0$	$y = 0$	$x = 0$	$x = 0$
Length of the latus rectum	4a	4a	4a	4a

Practice Problems on Parabola

Illustration 1: Find the vertex, axis, directrix, tangent at the vertex and the length of the latus rectum of the parabola

$$2y^2 + 3y - 4x - 3 = 0$$

.

Solution: The given equation can be re-written as

$$\left(y + \frac{3}{4}\right)^2 = 2\left(x + \frac{33}{32}\right)$$

which is of the form

$$Y^2 = 4aX$$

where

$$Y = y + \frac{3}{4}, X = x + \frac{33}{32}, 4a = 2$$

.

Hence, the vertex is $X = 0, Y = 0$, i.e. $(-33/32, -3/4)$.

The axis is

$$y + \frac{3}{4} = 0 \Rightarrow y = -\frac{3}{4}$$



$$y + \frac{3}{4} = 0 \Rightarrow y = -\frac{3}{4}$$

.

The directrix is $X = a = 0$.

$$\Rightarrow x + \frac{33}{32} + \frac{1}{2} = 0 \Rightarrow x = -\frac{49}{32}$$

The tangent at the vertex is

$$X = 0 \text{ or } x + \frac{33}{32} = 0 \Rightarrow x = -\frac{33}{32}$$

.

Length of the latus rectum $= 4a = 2$.

Illustration 2: Find the equation of the parabola whose focus is $(3, -4)$ and directrix $x - y + 5 = 0$.

Solution: Let $P(x, y)$ be any point on the parabola. Then,

$$\sqrt{(x - 3)^2 + (y + 4)^2} = \frac{|x - y + 5|}{\sqrt{1 + 1}}$$

$$\Rightarrow (x - 3)^2 + (y + 4)^2 = \frac{(x - y + 5)^2}{2}$$

$$\Rightarrow x^2 + y^2 + 2xy - 22x + 26y +$$

$$\Rightarrow (x + y)^2 = 22x - 26y - 25$$

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Illustration 2: Find the equation of the parabola whose focus is $(3, -4)$ and directrix $x - y + 5 = 0$.

Solution: Let $P(x, y)$ be any point on the parabola. Then,

$$\sqrt{(x-3)^2 + (y+4)^2} = \frac{|x-y+5|}{\sqrt{1+1}}$$

$$\Rightarrow (x-3)^2 + (y+4)^2 = \frac{(x-y+5)^2}{2}$$

$$\Rightarrow x^2 + y^2 + 2xy - 22x + 26y + 25 = 0$$

$$\Rightarrow (x+y)^2 = 22x - 26y - 25$$

Illustration 3: Find the equation of the parabola having focus $(-6, -6)$ and vertex $(-2, 2)$.

Solution: Let $S(-6, -6)$ be the focus and $A(-2, 2)$ the vertex of the parabola. On SA , take a point $K(x_1, y_1)$, such that $SA = AK$. Draw KM perpendicular on SK . Then, KM is the directrix of the parabola.



Since A bisects SK ,

$$\left(\frac{-6+x_1}{2}, \frac{-6+y_1}{2} \right) = (-2, 2)$$

$$\Rightarrow -6 + x_1 = -4 \text{ and } -6 + y_1 = 4 \text{ or } (x_1,$$

Hence, the equation

$$10 = m(x+2) \dots\dots(1)$$

Also gradient of

$$SK = \frac{10 - (-6)}{2 - (-6)} :$$



Illustration 3: Find the equation of the parabola having focus $(-6, -6)$ and vertex $(-2, 2)$.

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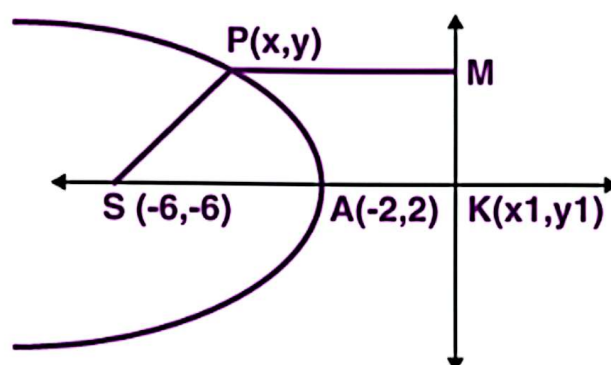
$$\Rightarrow -6 + x_1 = -4 \text{ and } -6 + y_1 = 4 \text{ or } (x_1,$$

Hence, the equation of the directrix KM is $y - 10 = m(x+2) \dots\dots(1)$

Also gradient of

$$SK = \frac{10 - (-6)}{2 - (-6)} = \frac{16}{8} = 2; m = \frac{-1}{2}$$

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So, equation (1) becomes

$$y - 10 = \frac{1}{2}(x - 2)$$

or

$$x + 2y - 22 = 0$$

is the directrix.

Next, let PM be perpendicular to the directrix
KM from any point P(x, y) on the parabola.

From $SP = PM$, the equation of the parabola
is

$$\sqrt{\{(x + 6)^2 + (y + 6)^2\}} = \frac{x + 2y - 22}{\sqrt{1^2 + 2^2}}$$

Illustration 4: Find the coordinates of the focus, the axis of the parabola, the equation of directrix and the length of the latus rectum for $y^2 = 12x$.

Solution: The given equation is $y^2 = 12x$.

Here, the coefficient of x is positive. Hence, the parabola opens towards the right.

On comparing this equation with $y^2 = 4ax$, we get $4a = 12$ or $a = 3$.

The coordinates of the focus are given by $(a, 0)$, i.e., $(3, 0)$.

Since the given equation involves y^2 , the axis of the parabola is the y -axis.

The equation of directrix is $x = -a$, i.e., $x = -3$

Length of latus rectum $= 4a = 4 \times 3 = 12$

Illustration 5: Find the coordinates of the focus, the axis of the parabola, the equation of directrix and the length of the latus rectum for $x^2 = -16y$.

Solution: The given equation is $x^2 = -16y$.

Here, the coefficient of y is negative.

Hence, the parabola opens downwards.

On comparing this equation with $x^2 = -4ay$, we get,

$$-4a = -16$$

$$a = 4$$

Coordinates of the focus = $(0, -a) = (0, -4)$.

Since the given equation involves x^2 , the axis of the parabola is the y -axis.

Equation of directrix, $y = a$, i.e., $= 4$

Length of latus rectum = $4a = 16$



Illustration 6: If the parabola $y^2 = 4x$ and $x^2 = 32y$ intersect at $(16, 8)$ at an angle θ , then find the value of θ .

Solution: The slope of the tangent to $y^2 = 4x$ at $(16, 8)$ is given by

$$m_1 = \left(\frac{dy}{dx} \right)_{(16,8)} = \left(\frac{4}{2y} \right)_{(16,8)} = \frac{2}{8} = \frac{1}{4}$$

The slope of the tangent to $x^2 = 32y$ at $(16, 8)$ is given by

$$m_2 = \left(\frac{dy}{dx} \right)_{(16,8)} = \left(\frac{2x}{32} \right)_{(16,8)} = 1$$

$$\therefore \tan \theta = \frac{1 - (1/4)}{1 + (1/4)} = \frac{3}{5}$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{3}{5} \right)$$

Illustration 7: Find the equation of the common tangent of $y^2 = 4ax$ and $x^2 = 4ay$.

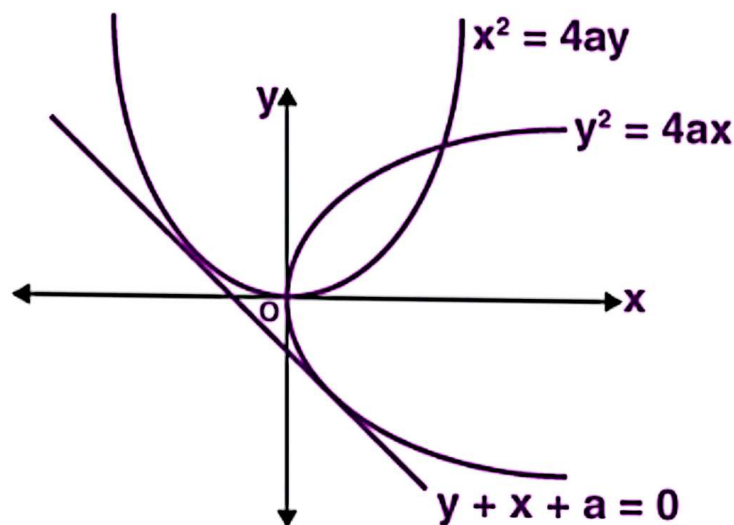
Solution: Equation of tangent to $y^2 = 4ax$ having slope m is

$$y = mx + \frac{a}{m}$$

.

It will touch $x^2 = 4ay$, if

$x^2 = 4a \left(mx + \frac{a}{m} \right)$ has equal roots.



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Thus,

$$16a^2m^2 = -16\frac{a^2}{m} \Rightarrow m = -1$$

Thus, the common tangent is $y + x + a = 0$.

Illustration 8: Find the equation of normal to the parabola $y^2 = 4x$ passing through the point $(15, 12)$.

Solution: The equation of the normal having slope m is

$$y = mx - 2m - m^3$$

If it passes through the point $(15, 12)$ then

$$12 = 15m - 2m - m^3$$

$$\Rightarrow m^3 - 13m + 12 = 0$$

$$\Rightarrow (m - 1)(m - 3)(m + 4) = 0$$

$$\Rightarrow m = 1, 3, -4$$

Hence, the equations of normal are:

$$y = x - 3, y = 3x - 33 \text{ and } y + 4x = 72$$

Illustration 9: Find the point on $y^2 = 8x$ where line $x + y = 6$ is normal.

Solution: Slope m of the normal $x + y = 6$ is -1 and $a = 2$

Normal to parabola at point $(am^2, -2am)$ is

$$y = mx - 2am - am^3$$

$$\Rightarrow y = -x + 4 + 2 \text{ at } (2, 4)$$

$$\Rightarrow x + y = 6 \text{ is normal at } (2, 4)$$

Illustration 10: Tangents are drawn to $y^2 = 4ax$ at a point where the line $lx + my + n = 0$ meets this parabola. Find the intersection of these tangents.

Solution: Let the tangents intersect at $P(h, k)$. Then, $lx + my + n = 0$ will be the chord of contact. That means $lx + my + n = 0$ and $yk - 2ax - 2ah = 0$, which is a chord of contact, will represent the same line.

Comparing the ratios of coefficients, we get

$$\frac{k}{m} = \frac{-2a}{l} = \frac{-2ah}{n}$$

$$\Rightarrow h = \frac{n}{l}, k = -\frac{2am}{l}$$



→ $x + y = 0$ is normal at $(2, 2)$

Illustration 10: Tangents are drawn to $y^2 = 4ax$ at a point where the line $lx + my + n = 0$ meets this parabola. Find the intersection of these tangents.

Solution: Let the tangents intersect at P (h, k). Then, $lx + my + n = 0$ will be the chord of contact. That means $lx + my + n = 0$ and $yk - 2ax - 2ah = 0$, which is a chord of contact, will represent the same line.

Comparing the ratios of coefficients, we get

$$\frac{k}{m} = \frac{-2a}{l} = \frac{-2ah}{n}$$

$$\Rightarrow h = \frac{n}{l}, k = -\frac{2am}{l}$$

Illustration 11: If the chord of contact of tangents from a point P to the parabola $y^2 = 4ax$ touches the parabola $x^2 = 4by$, then find the locus of P.

Solution: Chord of contact of parabola $y^2 = 4ax$ w.r.t. point P(x_1, y_1)

$$yy_1 = 2a(x + x_1) \dots (1)$$

This line touches the parabola $x^2 = 4by$.

Solving line (1) with parabola, we have

$$x^2 = 4b[(2a/y_1$$

or

$$y_1 x^2 - 8abx -$$

According to t



Illustration 11: If the chord of contact of tangents from a point P to the parabola $y^2 = 4ax$ touches the parabola $x^2 = 4by$, then find the locus of P.

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$$yy_1 = 2a(x + x_1) \dots\dots(1)$$

This line touches the parabola $x^2 = 4by$.

Solving line (1) with parabola, we have

$$x^2 = 4b \left[\left(\frac{2a}{y_1} \right) (x + x_1) \right]$$

or

$$y_1 x^2 - 8abx - 8abx_1 = 0$$

According to the question, this equation must have equal roots.

$$\Rightarrow D = 0$$

$$\Rightarrow 64a^2b^2 + 32abx_1y_1 = 0$$

$$\Rightarrow x_1y_1 = -2ab$$

or

$$xy = -2ab$$

, which is the required locus.



Illustration 12:

Length of intercept by the line $4y = 3x -$

(1) $\frac{9}{1600}$

(2) $\frac{1600}{9}$

(3) $\frac{160}{9}$

(4) None of these

Solution :

$$4y = 3x - 48 \Rightarrow m = 3/4, c = -12$$

$$y^2 = 64x \Rightarrow a = 16$$

$$\begin{aligned}\text{Length of intercept} &= \frac{4}{m^2} \sqrt{a(1+m^2)} \\ &= \frac{4}{9} \times 16 \sqrt{16 \left(1 + \frac{9}{16}\right)} \\ &= \frac{1600}{9}\end{aligned}$$

Answer : [2]

Illustration 13:

\ If two normals are drawn from any point

$$(1) y^2 = 4ax$$

$$(2) x^2 = 4ay$$

$$(3) y^2 = -4ax$$

$$(4) x^2 = -4ay$$

Solution :

\ Let the point is (h, k) .

The equation of any normal to the parabola

$$y = mx - 2am - am^3$$

Passes through (h, k)

$$k = mh - 2am - am^3$$

$$am^3 + m(2a - h) + k = 0$$

m_1, m_2, m_3 are roots of the equation

$$m_1 m_2 m_3 = -\frac{k}{a}$$

$$m_1 m_2 = 2, m_3 = -\frac{k}{2a}$$

m_3 is root of (i)

$$a\left(-\frac{k}{2a}\right)^3 - \frac{k}{2a}(2a - h) + k = 0 \Rightarrow k^2 = 4a(h - a)$$

Thus locus is $y^2 = 4a(x - a)$

Answer : [1]



Illustration 14:

\ If a normal chord of the parabola $y^2 =$

(1) ± 1

(2) $\pm \sqrt{2}$

(3) $\pm \sqrt{3}$

(4) None of these

Solution : If $P(at_1^2, 2at_1)$ be one end of th

Again slope of OP $= \frac{2at_1}{at_1^2} = \frac{2}{t_1}$

Slope of OQ $= \frac{2}{t_2} \quad \therefore \frac{2}{t_1} \times \frac{2}{t_2} = -1$

$\Rightarrow t_1 t_2 = -4$

From (1) and (2)

$-\frac{4}{t_1} = -t_1 - \frac{2}{t_1} \Rightarrow \frac{2}{t_1} = t_1 \Rightarrow t_1^2 = 2 \Rightarrow t_1 = \pm \sqrt{2}$

Answer : [2]

Frequently Asked Questions

Q1 What do you mean by the conjugate axis of a parabola?

The conjugate axis is the line perpendicular to the transverse axis of the parabola and passes through the vertex of the parabola.

Q2 What is the eccentricity of a parabola?

The eccentricity of a parabola equals 1.

Q3 Give the equation of the chord of contact of tangents from a point $P(x_1, y_1)$ to the parabola $y^2 = 4ax$.

The equation of the chord of contact of tangents from a point $P(x_1, y_1)$ to the parabola $y^2 = 4ax$ is given by $yy_1 - 2a(x + x_1) = 0$.

Q4 List two applications of a parabola.

Parabolic arch is used in architecture.

Parabolic reflectors are used in satellite dishes, reflecting telescopes, etc.

Q5 What do you mean by the foci of a parabola?

The focus of a parabola lies on the axis of the parabola. For the parabola having equation $y^2 = 4ax$ and having the x-axis as its axis, the focus is given by $F(a, 0)$.

Q6 What do you mean by the vertex of a parabola?

The vertex of a parabola is the point at the intersection of the parabola and the line of symmetry. It is the point where the parabola makes its sharpest turn. It will denote the maximum and minimum points of the parabola. For a parabola $y^2 = 4ax$, the vertex is $(0, 0)$.



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Q7 How is the graph of a parabola shaped?

The graph of a parabola is a U-shaped curve.

Q8 Give the formula for the length of the latus rectum of a parabola.

If $y^2 = 4ax$ denotes the parabola, then the length of the latus rectum is given by $4a$.